CLASS 9 · CBSE · GANITA MANJARI PART 2 · DPP SHEET

Chapter 9: Propositions and their Converses

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Write the converse of "If a number ends in 0, then it is divisible by 5". Is the converse true?
2. Give a counterexample to "If a number is divisible by 3, then it is divisible by 9".
3. "If a quadrilateral is a square, then its four sides are equal." Is this true? Is its converse true?
4. "If two angles are vertically opposite, then they are equal." Is the converse true?
5. "If n is even, then n^2 is even." Are the statement and its converse true?
6. Give a counterexample to "All numbers of the form $n^2 + n + 41$ are prime".
7. A triangle has sides 8 cm, 15 cm and 17 cm. Is it right-angled? Which result did you use?
8. "If a triangle is right-angled, then it has two acute angles." Is the converse true?
9. Are "If $ab = 0$, then $a = 0$" and its converse true?
10. "If n is divisible by 12, then it is divisible by 3 and by 4." Is the converse true?

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Answer Key

1. "If a number is divisible by 5, then it ends in 0." False: 15 is a counterexample.
2. 6 (or 12, 15, ...)
3. True. The converse is false: a rhombus has four equal sides but need not be a square.
4. No. The two base angles of an isosceles triangle are equal but are not vertically opposite.
5. Both are true.
6. $n = 40$ gives $1681 = 41^2$ (also $n = 41$)
7. Yes, since $64 + 225 = 289 = 17^2$; the converse of the Baudhāyana–Pythagoras theorem.
8. No. A triangle with angles 50°, 60° and 70° has two acute angles but is not right-angled.
9. The statement is false ($a = 3, b = 0$). The converse "If $a = 0$, then $ab = 0$" is true.
10. Yes, because 3 and 4 have no common factor.

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